Duc Nguyen, M.S.

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PROFESSIONAL SUMMARY

Software engineer with experience building Android marketplace features at Walmart Global Tech and backend, database, automation, and monitoring systems at BMW of North America. Experienced in **Kotlin**, **Java**, **Android**, **GraphQL**, cloud services, and **AI/ML** projects.

PROFESSIONAL EXPERIENCE

Walmart Global Tech - Android Developer

Sunnyvale, CA (Remote) | December 2023-Present

- Develop Walmart marketplace features in **Kotlin**, **Java**, and **GraphQL**, collaborating on complex integration and user experience.
- Build **LocalFinds** seller-discovery carousel and grid with delivery and pickup estimates and category tags; reused across **three pages**.
- Created a **LocalFinds** landing-page banner reused across **four pages**, with category navigation, promotional content, and an FAQ.
- Implement **LocalFinds** product fulfillment selection for shipping, pickup, and delivery, with fees, seller details, and return eligibility.
- Add **Walmart Resold** condition and seller-rating details to product pages and a condition-details bottom sheet.
- Implemented a search-tile savings checkbox that automatically applies claimed discounts in the cart.
- Build reusable UI components and contribute to Walmart's plugin-based, registry-driven **server-driven UI**.
- Build modular app architecture with **MVVM** and **multi-module Gradle** for reusable features.
- Fix memory leaks, image-loading problems, and lifecycle issues with **Android Profiler** and **LeakCanary**, improving app stability and responsiveness.
- Prevent large-list bottom-sheet crashes by chunk-loading items while backend pagination was unavailable.
- Establish automated testing with **JUnit**, **YAML UI automation**, **Sauce Labs**, and **Jenkins CI**.
- Stabilize CI by rewriting flaky, timing-out tests and removing unnecessary mocks, logic, and waits.
- Cut **YAML** automation runtime by about **45 minutes per CI job** through sharding and trimming.
- Partner with UX/UI designers to redesign app screens around user needs.
- Review code and mentor about three Android engineers, including rotating team members and contractors.

BMW of North America LLC - Software Developer Intern

New Jersey, USA | September 2022-March 2023

- Migrated AWS Lambda services for the dealer part-request warranty-claim system from **Spring MVC** to **Quarkus**, reducing cold-start time by **50%**.
- Migrated a mainframe DB2 database to **Oracle SQL** and **Salesforce**, saving **$50,000 in annual API costs** and improving data access.
- Optimized a **UiPath RPA** bot to remove manual work, generating **$500,000 in annual labor savings**.
- Built an **Angular** and **Spring MVC** monitoring service that polls AWS Lambda functions and reports pass/fail status and run duration to senior engineers and the engineering manager.
- Implemented **Jenkins CI/CD** and updated tests to track code coverage and test pass rates, improving system reliability.

PROJECTS

EasyChef - AI-Powered Recipe Generator

Technologies: Python, **Flask**, **TensorFlow**, **Hugging Face**

- Developed a web-based AI recipe generator that processes user input; explored efficient model deployment for responsive user-facing applications.

Movie Recommendation System - Scalable Media Platform

Technologies: PySpark, **Hadoop**, **Flask**, **CherryPy**

- Built a movie recommendation system to process large media metadata sets and provide timely, personalized recommendations.

EDUCATION

Rutgers University

Master of Science in Software Engineering | May 2023

Rutgers University

Bachelor of Science in Computer Engineering | May 2021

TECHNICAL SKILLS

Languages: Kotlin, Java, Python, JavaScript, C++, C, Scala, R

Mobile: Kotlin, Java, Android SDK, XML layouts, Coroutines, MVVM, multi-module Gradle

Backend and APIs: Spring MVC, Quarkus, Node.js, Flask, GraphQL, Apollo

Frontend: React, Angular, JavaScript, HTML, CSS

Data and databases: MySQL, MongoDB, Firebase, Oracle SQL, DB2, Hadoop, PySpark

AI and machine learning: TensorFlow, Hugging Face

Cloud and DevOps: AWS Lambda, AWS S3, Jenkins, Git, GitHub, Linux

Testing and performance: JUnit 5, YAML-based UI automation, Sauce Labs, Android Profiler, LeakCanary

Architecture: Clean architecture, server-driven UI, plugin-based architecture, modularization